

Dual-Branch ResNet50 — Ablation Study

4-Model Comparison | model.ipynb | 2026

1 Study Overview

#	Model	Fusion Strategy	Attn output?
1	Baseline	Concat → Conv1×1 → AvgPool	—
2	CrossAttn	Unidirectional cross-attention (Q=clinic, K,V=derm)	✓ [B,49,49]
3	ChannelAttn	SE channel attention per branch → Concat	—
4	Cross+ChannelAttn	Channel attention → Cross-attention	✓ [B,49,49]

All four models share the same backbone, dataset, splits, and training protocol. Only the **fusion mechanism** differs — making this a controlled ablation study.

2 Shared Backbone

Clinical photo → ResNet50_A → [B, 2048, 7, 7] ■■
■→ [FUSION] → Classifier → 5 classes
Dermoscopic → ResNet50_B → [B, 2048, 7, 7] ■■

Component	Detail
Backbone (×2)	ResNet50, ImageNet pretrained (frozen layers 1–3 during warm-up)
Spatial tokens	7×7 feature map → 49 tokens per branch
Freeze schedule	Epochs 1–5: layers 1–3 frozen, layer4 trainable
Unfreeze	Epoch 6+: layer3+4 unfrozen, lr halved
Classifier head	Varies by model (see Section 3)
Output	5-class logits (BCC, MEL, MISC, NV, SK)

3 Model Architectures

Model 1 — Baseline (no attention)

feat_c [B,2048,7,7] + feat_d [B,2048,7,7] → Cat [B,4096,7,7]
→ Conv2d(4096→1024, 1×1) → BN → ReLU → AvgPool → [B,1024]
→ Linear(1024→512) → BN → ReLU → Dropout → Linear(512→5)

Model 2 — CrossAttn (unidirectional cross-attention)

Q ← Linear(2048→256) on clinic tokens [B, 49, 256]
K,V ← Linear(2048→256) on derm tokens [B, 49, 256]
MultiheadAttention(num_heads=4) → Residual+LN → FFN(GELU) → Residual+LN
→ AvgPool → [B, 256] → Linear(256→128) → BN → ReLU → Dropout → Linear(128→5)

Model 3 — ChannelAttn (SE channel attention per branch)

feat_c → ChannelAttention(2048, reduction=16) → scaled feat_c
feat_d → ChannelAttention(2048, reduction=16) → scaled feat_d
Cat [B,4096,7,7] → Conv1×1(1024) → BN → ReLU → AvgPool

→ Linear(1024→512) → BN → ReLU → Dropout → Linear(512→5)

Model 4 — Cross+ChannelAttn (channel attention → cross-attention)

feat_c → ChannelAttention → refined_c [B, 2048, 7, 7]
feat_d → ChannelAttention → refined_d [B, 2048, 7, 7]
Q ← refined_c, K,V ← refined_d → CrossAttentionFusion → [B, 256]
→ Linear(256→128) → BN → ReLU → Dropout → Linear(128→5)

SE Block (ChannelAttention)	Detail
Avg-pool path	AdaptiveAvgPool2d(1) → FC(2048→128→2048)
Max-pool path	AdaptiveMaxPool2d(1) → FC(2048→128→2048) (shared weights)
Scale	Sigmoid(avg_out + max_out) → multiply feature map channel-wise
Reduction ratio	16 (2048 → 128 → 2048)

4 Fixed Hyperparameters (all models identical)

Parameter	Value	Applies to
Learning rate (warm-up)	0.000750	All models
Learning rate (after unfreeze)	0.000375 (lr / 2)	All models
Dropout	0.32	All models
embed_dim	256	CrossAttn, Cross+ChannelAttn
num_heads	4	CrossAttn, Cross+ChannelAttn
weight_decay	0.000957	All models
batch_size	64	All models
Max epochs	30	All models
Early stopping patience	7 (monitors val_loss)	All models
Unfreeze epoch	6 (after epoch 5)	All models
LR scheduler	StepLR(step=10, γ=0.5)	All models
Loss function	CrossEntropyLoss with inverse class weights	All models
Optimizer	Adam	All models
Seed	42	All models

5 Training Protocol

Phase	Epochs	Layers trainable	LR
Warm-up	1–5	layer4 + fusion + classifier (layers 1–3 frozen)	0.000750
Full training	6–30	layer3 + layer4 + fusion + classifier	0.000375

Early stopping restores best-val-loss weights automatically. Each model is trained independently from scratch with a fresh optimizer and scheduler.

6 Dataset & Splits

Source	Samples	Type
Derm7pt	1,011	Paired clinical + dermoscopic images
MILK10k	4,361	Paired clinical + dermoscopic images
Combined	5,372	Stratified 70 / 15 / 15 split

Split	Samples
Train	3,760
Validation	806
Test	806

Class	Description	Loss Weight
BCC	Basal Cell Carcinoma	0.419
MEL	Melanoma (all subtypes)	1.532
MISC	Lentigo, DF, Vascular, etc.	5.489
NV	Nevus (all subtypes)	0.813
SK	Seborrheic Keratosis	1.825

Class imbalance handled via **WeightedRandomSampler** (training batches) and **weighted CrossEntropyLoss** (inverse class frequency).

7 Saved Artifacts (per run)

File	Contents
model_baseline.pth	Baseline weights + history + test metrics
model_cross_attn.pth	CrossAttn weights + history + test metrics
model_channel_attn.pth	ChannelAttn weights + history + test metrics
model_cross_channel_attn.pth	Cross+ChannelAttn weights + history + test metrics
all_histories.pkl	Train/val loss + accuracy per epoch for all 4 models
all_metrics.pkl	Test metrics dict for all 4 models
all_eval_data.pkl	y_true, y_pred, y_probs, attn weights for all 4 models
comparison_metrics.csv	Side-by-side metrics table (CSV)

8 Comparison Visualizations (Section 10)

Plot	Description
10.1 Training curves	Loss + accuracy per epoch; solid=val, dashed=train; 4 colors
10.2 Metrics bar + F1 heatmap	Grouped bar chart (6 metrics) + per-class F1 heatmap (4×5)

10.3 ROC curves	One subplot per class (5 total), 4 lines per model
10.4 Confusion matrices	2×2 grid of normalized confusion matrices (one per model)
10.5 Cross-attention maps	Attention overlay on dermoscopic images (CrossAttn & Cross+Channel only)
10.6 Sample predictions	Correct vs incorrect examples from CrossAttn model